Introduction to AI and Big Data in Finance

Project Tutorial: Baseline Pipeline and Practical Tips

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Project overview

What this project is really about

- Build a full machine learning pipeline for **cross-sectional stock return prediction**
- Use the JKP dataset to go from:
  - Features
  - Predicted returns
  - Portfolio weights
  - Evaluation metrics
- What matters most:
  - Sound methodology
  - **No look-ahead bias**
  - **Economic reasoning**
- Realistic results are often modest
  - Suspiciously strong results are often a red flag for leakage or benchmark mistakes

The big-picture pipeline

- Choose predictors
- Preprocess carefully
- Fit models on the training period
- Tune hyperparameters on the validation period
- Refit in an expanding window through the test period
- Convert predicted returns into portfolio weights
- Evaluate with:
  - Out-of-sample R^2
  - Sharpe ratio with annualization
  - Information coefficient (optional)

Data and split

What exactly is the prediction problem?

- Each observation is a **stock-month**
- Predictors are characteristics observed at the end of month t
- The target is next month's excess stock return:

```
ret_exc_lead1m
```

- This is a **cross-sectional** prediction problem:
  - We rank stocks *within each month*
  - We care about relative ordering across stocks, not just level prediction
- This is different from market timing:
  - We are not predicting one return for the whole market
  - We are predicting many stock-level returns each month

Train-validation-test split

- Use a **time-based split**, not a random split
- Recommended windows:
  - **Training**: January 2005 – December 2015
  - **Validation**: January 2016 – December 2018
  - **Test**: January 2019 – December 2024
- Roles of each window:
  - Training: fit preprocessing and model parameters
  - Validation: choose hyperparameters
  - Test: final out-of-sample evaluation only
- Key rules:
  - No random shuffling of stock-month observations
  - **No tuning on the test set**
  - Keep the long recent test period intact

Preprocessing without look-ahead bias

- Typical preprocessing steps:
  - Log-transform *skewed features* (not all features)
  - Winsorize outliers
  - Impute missing values
  - Standardize predictors
- Fit preprocessing parameters on **training data only**
- Apply those same fitted transformations to validation and test data
- In expanding-window retraining:
  - Refit preprocessing only using data available up to that date
- Leakage often enters through preprocessing
  - E.g., computing medians or standard deviations using the full sample

Baseline models

Required baseline models

- A good baseline model set includes:
  1. Historical-average benchmark
  2. OLS baseline
  3. One ML model such as Ridge, Lasso, Elastic Net, or another justified model
- Good practice:
  - Start with the same cleaned feature set across models (no cherry-picking)
  - Explain every hyperparameter choice
- A very useful comparison:
  - OLS vs. Ridge on the same predictors
  - This isolates the value of regularization
- The goal is not just to run models:
  - The goal is to justify why one specification works better than another (if it does)

Important benchmark distinction

- There are **two different benchmark ideas** that students should not confuse
- **Historical-average model for stock i:**
  - Predict stock i using its own past average return
- **Null forecast for out-of-sample R^2:**
  - Use a common unconditional mean benchmark
- These two benchmarks are not necessarily the same
- If you use both, define them clearly in the report and presentation
- The notebook simplification and the project handout may not be perfectly identical
  - If needed, implement the stock-specific historical average separately

Tuning and expanding-window retraining

- Use the validation period to choose hyperparameters
- Example:
  - Choose Ridge penalty α (or λ) from a grid of values
- Once a hyperparameter is chosen, **fix it** before moving to the test period
- In the test period:
  - Refit preprocessing and the model in an expanding window (optional)
  - Use all data available up to that time (this usually improves performance)
  - A common implementation is to refit every January and predict the next year
- Why this matters:
  - A model fit only through 2015 may become stale by 2024

Portfolio construction and evaluation

From predicted returns to portfolio weights

- Each month, the model produces predicted return $\hat{r}_{i,t+1}$ for each stock i
- Convert predictions into a cross-sectional **signal** and **weight**
- **Default method** (no need to use this exact method):
  - Rank predicted returns within each month in descending order
  - Center the monthly ranks by subtracting their cross-sectional mean
  - Normalize the centered ranks so that gross exposure equals 2
  - This yields a **zero-cost long-short portfolio**
- Interpretation:
  - Stocks with higher predicted returns receive positive weights
  - Stocks with lower predicted returns receive negative weights
  - But the weighting is not too extreme
- Optional modifications:
  - Cap individual positions, set some weights to zero, etc.

From weights to realized returns and Sharpe ratio

- Monthly portfolio return:

$$r_{p,t+1} = \sum_i w_{i,t} r_{i,t+1}$$

- Annualized mean return:

$$\mu_{\text{ann}} = 12\bar{r}_m$$

- Annualized volatility:

$$\sigma_{\text{ann}} = \sqrt{12}\sigma_m$$

- Sharpe ratio:

$$\text{SR} = \frac{\mu_{\text{ann}}}{\sigma_{\text{ann}}}$$

- Because our returns are already excess returns, the above is fine

Out-of-sample R^2, IC, and why they can disagree

- Out-of-sample R^2 measures prediction error relative to a benchmark:

$$R^2_{\text{OOS}} = 1 - \frac{\text{MSE}_{\text{model}}}{\text{MSE}_{\text{null}}}$$

- IC (optional) measures ranking quality:
  - Usually the monthly Spearman rank correlation between predicted and realized returns
- Key difference:
  - R^2 focuses on **level accuracy**
  - IC and Sharpe often depend more on **ranking quality**
- Important practical point:
  - A model can have weak or negative R^2_{OOS}
  - Yet still produce useful rankings and a reasonable Sharpe ratio

Pitfalls and presentation

Common mistakes

- **Incorrect split**:
  - Random train/test split instead of date-based split
- **Leakage in preprocessing**:
  - Fitting scaler, imputer, or winsorization bounds on the full sample
- **Improper tuning**:
  - Choosing hyperparameters using the test period
- **Walk-forward mistakes**:
  - Forgetting expanding-window refits
- **Portfolio construction mistakes**:
  - Using realized returns directly in signal construction
  - Comparing long-only and long-short portfolios without explanation
- **Economic realism**:
  - Ignoring turnover, liquidity, and transaction-cost discussion

What a strong project and presentation looks like

- Get the baseline working **end to end**
  - Preprocessing
  - Model fitting
  - Walk-forward prediction
  - Portfolio construction
  - Evaluation
- Then add one meaningful exploration direction
- Show a few clear outputs:
  - Model comparison table
  - Cumulative return figure
  - Feature importance or sensitivity analysis (optional)
- Interpret results economically
- Explain the pipeline and findings, but do not walk through code line by line!

Appendix

Appendix: Clarifying the historical-average benchmark

- The handout-style benchmark is:
  - A **stock-specific** historical average
  - Each stock is predicted using its own past mean return
- The Campbell–Thompson null for R^2_{OOS} is often:
  - A **common unconditional mean** forecast
- These serve different purposes:
  - One is a benchmark prediction model
  - One is a benchmark for forecast evaluation
- If you implement both:
  - Label them clearly
  - Explain the distinction in the report